5-Bay SATA Drive Power Supply

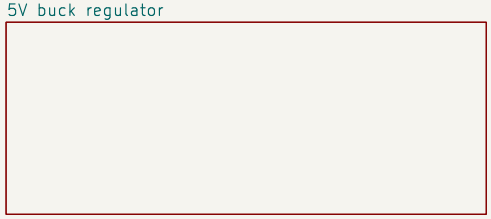
Single 12V brick, onboard 5V buck, five analog-staggered channels

Input protection



File: sata_psu_input.kicad_sch

5V buck regulator



File: sata_psu_buck.kicad_sch

Channel 1



File: sata_psu_channel1.kicad_sch

Channel 2



File: sata_psu_channel2.kicad_sch

Channel 3



File: sata_psu_channel3.kicad_sch

Channel 4



File: sata_psu_channel4.kicad_sch

Channel 5



File: sata_psu_channel5.kicad_sch

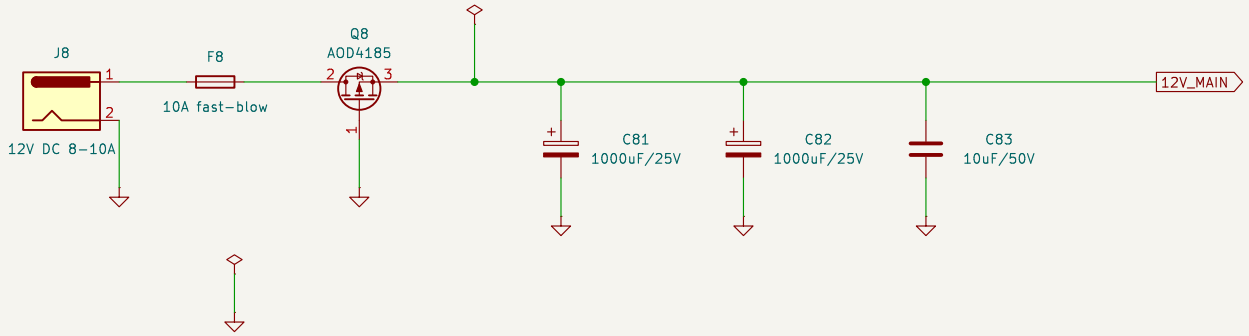
5-Bay SATA Drive PSU

Sheet: /
File: sata_psu.kicad_sch

Title: 5-Bay SATA Drive Power Supply

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 1/8

Input protection — 12V brick, fuse, reverse-polarity guard, bulk C



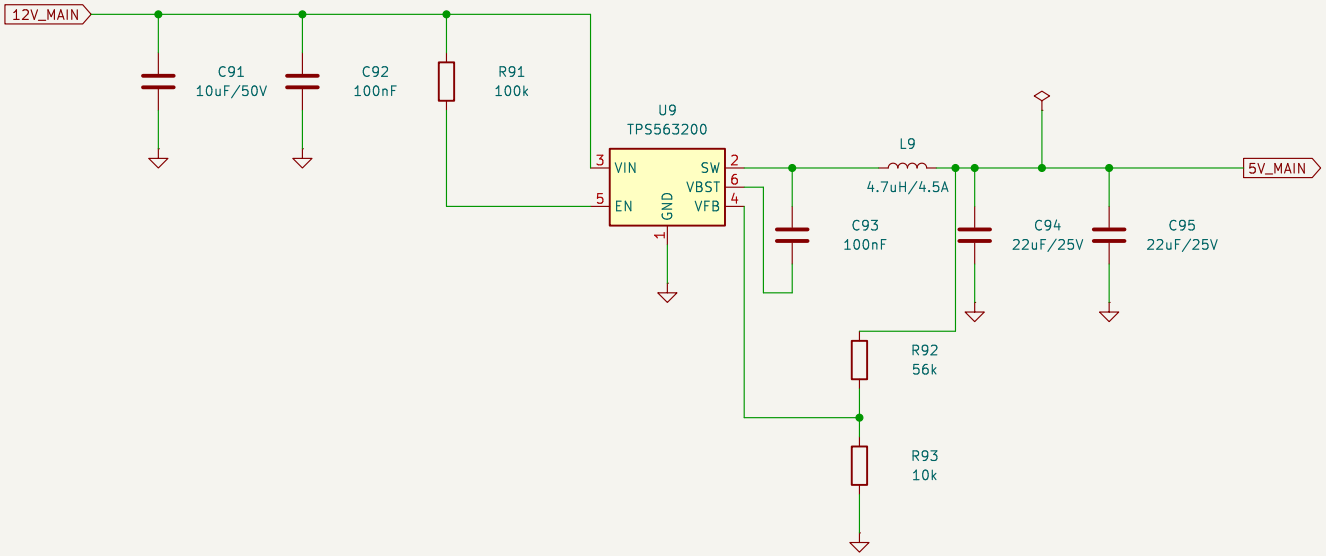
Star ground: single point back to the brick negative

5-Bay SATA Drive PSU		
Sheet: /Input protection/ File: sata_psu_input.kicad_sch		
Title: Input protection		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 2/8

12V to 5V buck (always on, feeds all five bays)

Vout = 0.768V x (1 + R92/R93) = 0.768 x (1 + 56k/10k) = 5.07V

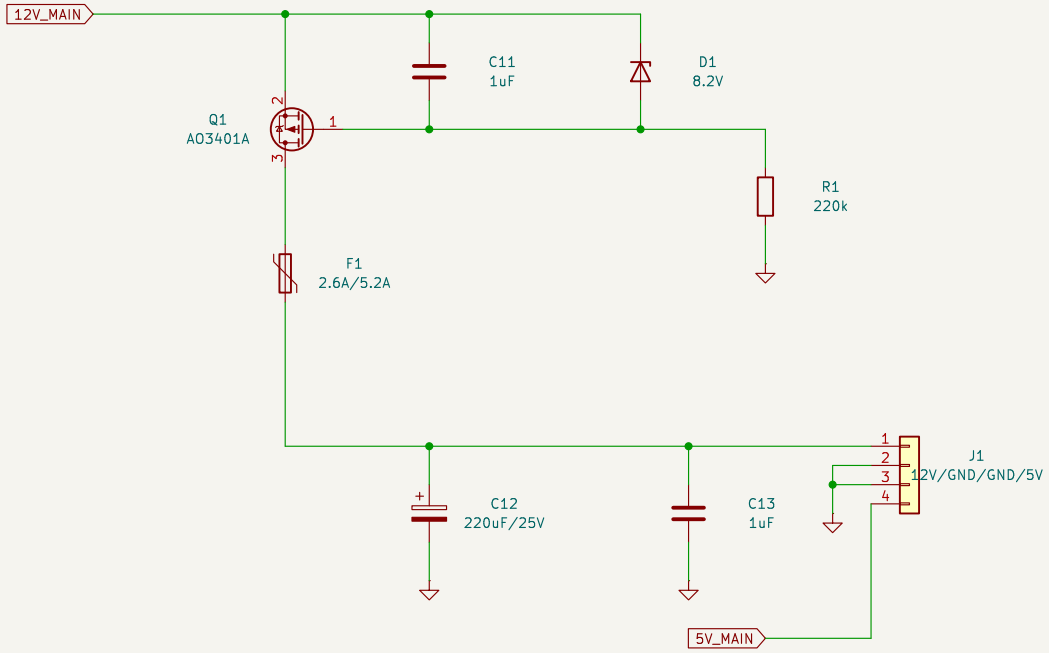
WARNING: TPS563200 is a 3A part; doc section 2 targets a 5A 5V rail.



5-Bay SATA Drive PSU		
Sheet: /5V buck regulator/		
File: sata_psu_buck.kicad_sch		
Title: 5V buck regulator		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 3/8

Channel 1 — staggered high-side switch

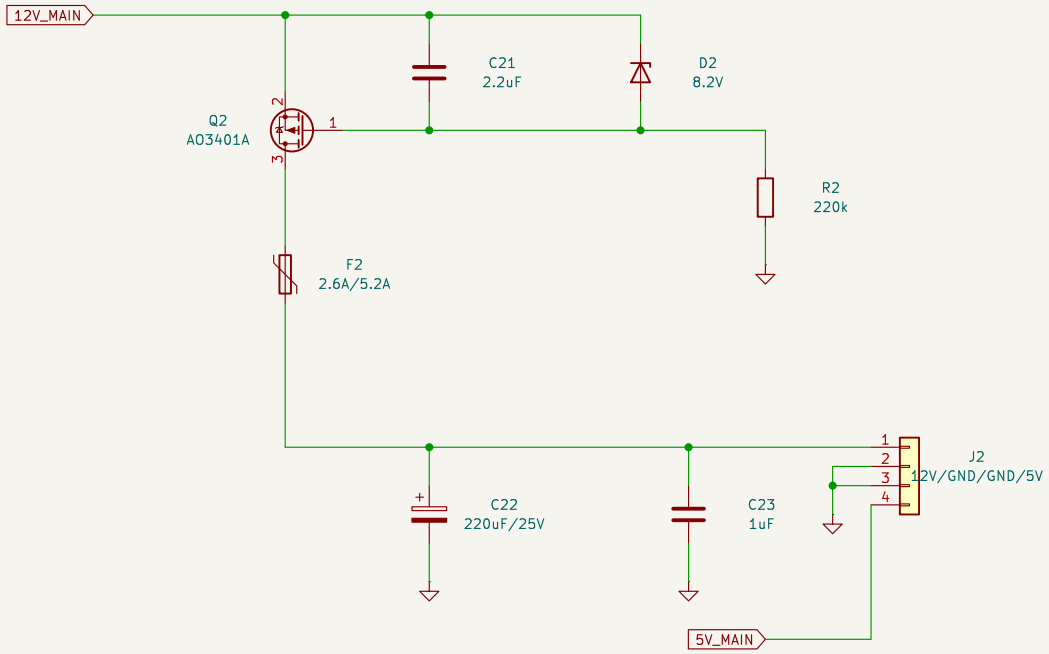
C11=1uF with R1=220k sets turn-on at ~0.2s
D1 clamps Vgs to 8.2V; AO3401A is rated +/-12V



5-Bay SATA Drive PSU		
Sheet: /Channel 1/ File: sata_psu_channel1.kicad_sch		
Title: Channel 1		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 4/8

Channel 2 — staggered high-side switch

C21=2.2uF with R2=220k sets turn-on at ~0.5s
D2 clamps Vgs to 8.2V; AO3401A is rated +/-12V

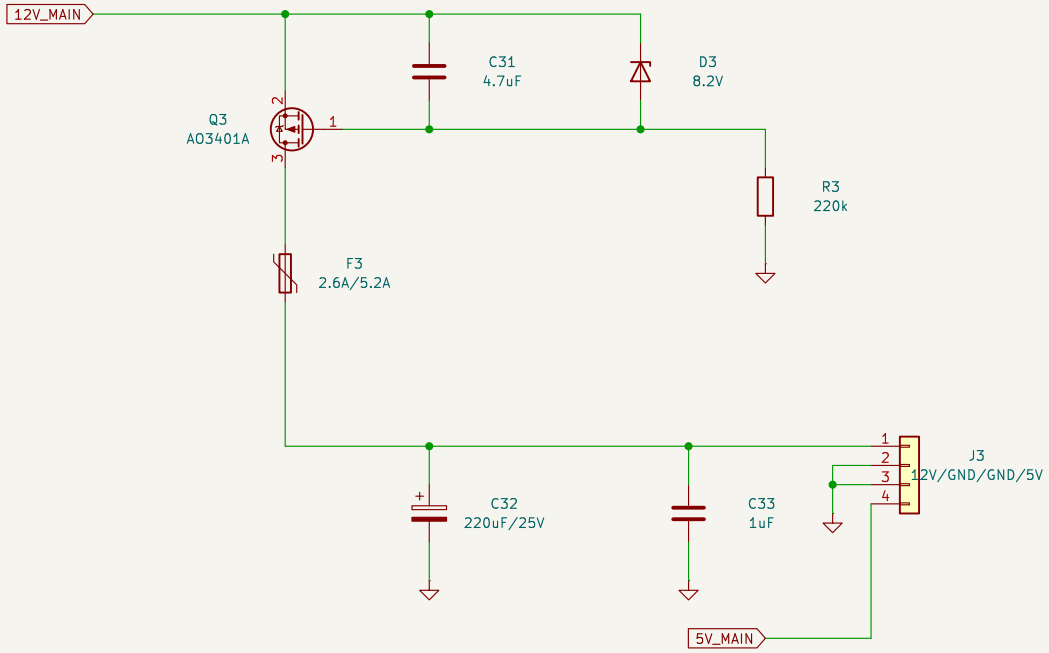


5-Bay SATA Drive PSU	
Sheet: /Channel 2/	
File: sata_psu_channel2.kicad_sch	
Title: Channel 2	
Size: A4	Date:
KiCad E.D.A. 9.0.3	Id: 5/8

Channel 3 — staggered high-side switch

C31=4.7uF with R3=220k sets turn-on at ~1s

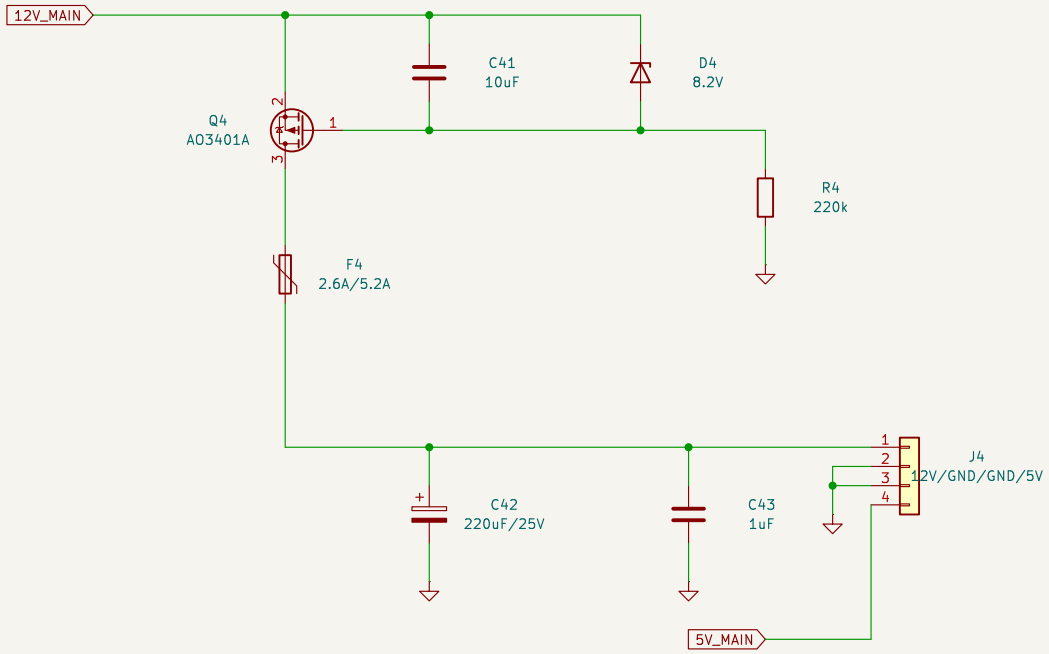
D3 clamps Vgs to 8.2V; AO3401A is rated +/-12V



5-Bay SATA Drive PSU	
Sheet: /Channel 3/	
File: sata_psu_channel3.kicad_sch	
Title: Channel 3	
Size: A4	Date:
KiCad E.D.A. 9.0.3	Id: 6/8

Channel 4 — staggered high-side switch

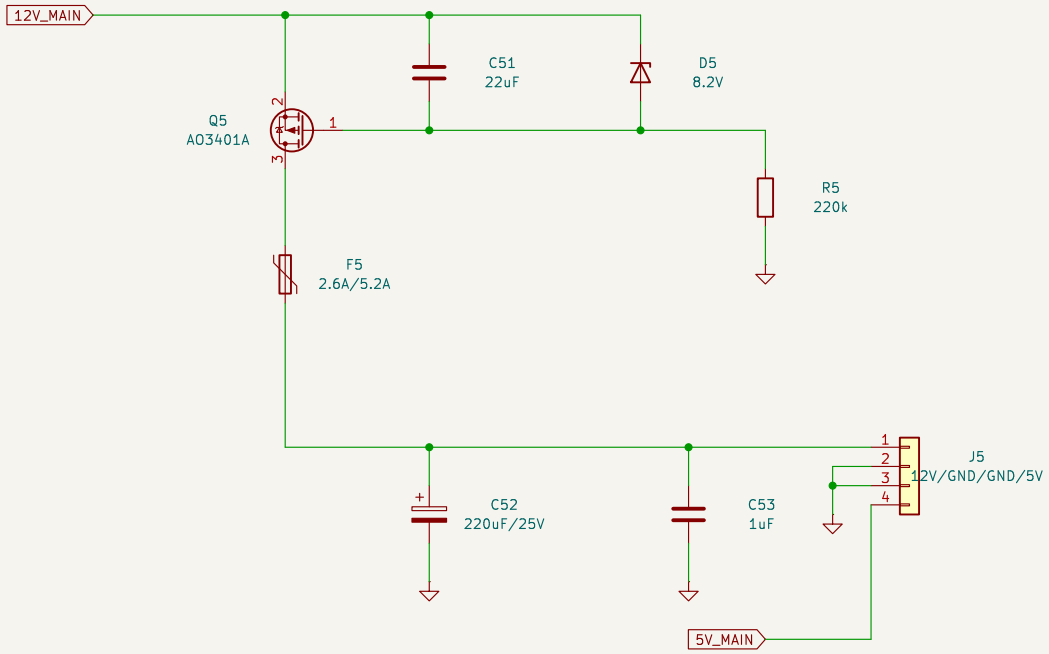
C41=10uF with R4=220k sets turn-on at ~2.2s
D4 clamps Vgs to 8.2V; AO3401A is rated +/-12V



5-Bay SATA Drive PSU		
Sheet: /Channel 4/		
File: sata_psu_channel4.kicad_sch		
Title: Channel 4		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 7/8

Channel 5 — staggered high-side switch

C51=22uF with R5=220k sets turn-on at ~4.8s
D5 clamps Vgs to 8.2V; AO3401A is rated +/-12V



5-Bay SATA Drive PSU		
Sheet: /Channel 5/		
File: sata_psu_channel5.kicad_sch		
Title: Channel 5		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 8/8